Math 220B HW 2

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1 D

Problem 1

Let R be a semilocal ring and let M and N be two R-modules such that $M \otimes_R N = 0$. Prove or disprove that either $M = 0$ or $N = 0$.

Solution: We'll disprove the statement by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, which it is semilocal ring because it only has finitely many maximal ideals (in fact, finitely many ideals since the ring is finite).

Now, consider the canonical projection $\pi_2 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\pi_3 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$, notice that since $6\mathbb{Z} = 2\mathbb{Z} \cap 3\mathbb{Z}$, one has the ideal $6\mathbb{Z} \subset \ker(\pi_2) = 2\mathbb{Z}$ and $6\mathbb{Z} \subset \ker(\pi_3) = 3\mathbb{Z}$. Hence, by Generalized First Isomorphism Theorem, with the canonical projection $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, the above two maps uniquely factor into surjective maps $\varphi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\varphi_3 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ as follow:

$$\begin{tikzcd} \mathbb{Z} \arrow[rr, "\pi_2"] \arrow[dr, "\pi_6"'] & & \mathbb{Z}/2\mathbb{Z} \\ & \mathbb{Z}/6\mathbb{Z} \arrow[ur, "\varphi_2"'] & \end{tikzcd} \qquad \begin{tikzcd} \mathbb{Z} \arrow[rr, "\pi_3"] \arrow[dr, "\pi_6"'] & & \mathbb{Z}/3\mathbb{Z} \\ & \mathbb{Z}/6\mathbb{Z} \arrow[ur, "\varphi_3"'] & \end{tikzcd}$$

Hence, $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$ can be realized as R-module, via the map φ_2 and φ_3 (in particular, one has $\varphi_2(\overline{1}_6) = \overline{1}_2$ and $\varphi_3(\overline{1}_6) = \overline{1}_3$ as the map, based on the above commutative diagrams).

Now, consider the tensor $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z}$, we claim that it is 0: Recall that $2, 3 \in \mathbb{Z}$ are coprime, with Bezout's Lemma there exists integers $s, t \in \mathbb{Z}$, such that $1 = 2s + 3t$. Then, under the projection map $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, one knows there exists $\overline{s}_6, \overline{t}_6 \in \mathbb{Z}/6\mathbb{Z}$, such that the following holds:

$$\overline{1}_6 = \overline{2}_6 \cdot \overline{s}_6 + \overline{3}_6 \cdot \overline{t}_6 \tag{1.1}$$

Then, for any $\overline{a}_2 \in \mathbb{Z}/2\mathbb{Z}$ and $\overline{b}_3 \in \mathbb{Z}/3\mathbb{Z}$, one has the following:

$$\begin{aligned} \overline{a}_2 \otimes \overline{b}_3 &= \overline{1}_6 \cdot \left(\overline{a}_2 \otimes \overline{b}_3\right) = (\overline{2}_6 \cdot \overline{s}_6 + \overline{3}_6 \cdot \overline{t}_6) \cdot \left(\overline{a}_2 \otimes \overline{b}_3\right) \\ &= \overline{s}_6 \cdot \left((\overline{2}_6 \cdot \overline{a}_2) \otimes \overline{b}_3\right) + \overline{t}_6 \cdot \left(\overline{a}_2 \otimes \left(\overline{3}_6 \cdot \overline{b}_3\right)\right) \\ &= \overline{s}_6 \cdot \left((\overline{0}_2 \overline{a}_2) \otimes \overline{b}_3\right) + \overline{t}_6 \cdot \left(\overline{a}_2 \otimes \left(\overline{0}_3 \overline{b}_3\right)\right) \\ &= \overline{s}_6 \cdot \left(\overline{0}_2 \otimes \overline{b}_3\right) + \overline{t}_6 \cdot \left(\overline{a}_2 \otimes \overline{0}_3\right) \\ &= 0 \end{aligned} \tag{1.2}$$

(Note: By definition, one has $\overline{2}_6 \cdot \overline{a}_2 := \varphi_2(\overline{2}_6)\overline{a}_2 = \overline{0}_2\overline{a}_2 = \overline{0}_2$, and same idea for $\mathbb{Z}/3\mathbb{Z}$).

Hence, this proves that $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z} = 0$. Yet, each $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z} \neq 0$ as R-module, which is a desired counterexample for the problem.

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Problem 2

Let $0 \to M' \xrightarrow{u} M \xrightarrow{v} M'' \to 0$ be an exact sequence of R-modules. Assume that M' and M'' are finitely generated R-modules. Prove that M is a finitely generated R-modules. Explain if the converse is true.

Solution: We'll first prove the given statement, then prove the converse is false in general, by providing a counterexample.

I. Proof of Statement:

Given the given sequence is exact, then it implies $M' \xrightarrow{u} M$ is injective, hence $M' \cong \text{im}(u) \subset M$, WLOG we'll denote $\text{im}(u)$ as M', and $M' \xrightarrow{u} M$ as an inclusion map, and recognize all elements in M' as in M also.

Since M', M'' are finitely-generated, there exists $x_1, \ldots, x_n \in M'$ and $y_1, \ldots, y_m \in M''$, such that $M' = \sum_{i=1}^n Rx_i$ and $M'' = \sum_{j=1}^m Ry_j$. Then, choose $z_1, \ldots, z_m \in M$, such that for each index $j \in \{1, \ldots, m\}$, one has $v(z_j) = y_j$. We claim that the collection $\{x_1, \ldots, x_n, z_1, \ldots, z_m\}$ generates M.

For any $x \in M$, since $v(x) \in M''$, there exists $b_1, \ldots, b_m \in R$, such that the following holds:

$$v(x) = \sum_{j=1}^m b_j y_j = \sum_{j=1}^m b_j v(z_j) = v\left(\sum_{j=1}^m b_j z_j\right) \tag{2.1}$$

As a result, one has $v\left(x - \sum_{j=1}^m b_j z_j\right) = 0$, or $x - \sum_{j=1}^m b_j z_j \in \ker(v)$.

Now, using exactness of the sequence, one gets $\ker(v) = \text{im}(u) = M'$, hence there exists $a_1, \ldots, a_n \in R$, such that $x - \sum_{j=1}^m b_j z_j = \sum_{i=1}^n a_i x_i$ by the finitely generated property of M'.

As a result, we have $x = \sum_{i=1}^n a_i x_i + \sum_{j=1}^m b_j z_j$, hence all element $x \in M$ can be expressed with some R-linear combination of $\{x_1, \ldots, x_n, z_1, \ldots, z_m\}$, showing M is finitely generated.

II. Counterexample of the Converse:

Let k be a field, consider the ring $R = k[x_1, x_2, \ldots]$, which R is a polynomial ring over k, with (countably) infinite indeterminates. First, let's note that R consists of polynomials that are finite sums of monomials, and each monomial can involve only finite indeterminates.

Consider the ideal $I := (x_1, x_2, \ldots)$ (ideal generated by all indeterminates). Then, $I, R, R/I$ are all R-modules, in particular R is a finitely generated R-module (since every $r \in R$ can be written as $r \cdot 1$, so R is generated by 1 as an R-module). Moreover, we naturally have the following exact sequence of R-linear maps:

$$0 \to I \hookrightarrow R \twoheadrightarrow R/I \to 0$$

And it's simply because the image of the inclusion $I \hookrightarrow R$ is I, and the kernel of projection $R \twoheadrightarrow R/I$ is also I.

Now, we have a short exact sequence, such that the middle R-module is finitely generated. Yet, we claim that I is not a finitely generated R-module:

Suppose the contrary that I is a finitely generated R-module, then there exists $f_1, ..., f_n \in I$, such that $I = \sum_{i=1}^{n} Rf_i = (f_1, ..., f_n)$ as ideal. Based on our definition, there are only finitely many indeterminates involved in $f_1, ..., f_n$, say $S = \left\{x_{i_1}, ..., x_{i_k}\right\}$ (and say it's ordered such that $i_1 < ... < i_k$).

Then, consider the element $x_{i_k+1} \in I$: By our assumption, there exists polynomials $g_1, ..., g_n \in R$, such that $x_{i_k+1} = \sum_{i=1}^{n} g_i f_i$. Which, let $T = \left\{x_{j_1}, ..., x_{j_l}\right\}$ be the (finite) indeterminates involved in the polynomials $g_1, ..., g_n$. Which, if consider the set of indeterminates $S \cup T \cup \left\{x_{i_k+1}\right\} = \left\{x_{i_k+1}, x_{m_1}, ..., x_{m_p}\right\}$ after reordering, one can realize the elements $x_{i_k+1}, f_1, ..., f_n, g_1, ..., g_n$ as elements in $k\left[x_{i_k+1}, x_{m_1}, ..., x_{m_p}\right] \hookrightarrow k[x_1, x_2, ...]$, and inside this smaller subring, the equality $x_{i_k+1} = \sum_{i=1}^{n} g_i f_i$ still holds.

Now, consider the evaluation map $k\left[x_{i_k+1}, x_{m_1}, ..., x_{m_p}\right] \to k$, by $x_{i_k+1} \mapsto 1$, and all $x_{m_r} \mapsto 0$: Since we've chosen x_{i_k+1}, such that it's distinct from all x_{i_j} $(1 \leq j \leq k)$, then one has x_{i_j} being some of the x_{m_r}, hence $x_{i_j} \mapsto 0$. Also, notice that since each $f_i \in I$, then it has no constant term (since one has $f_i = h_1 x_1 + ...$ for $h_i \in R$, such that finitely many $h_i \neq 0$; then, all the constant terms in h_i are multiplied with x_i, which is no longer with constant term). Hence, with each $x_{i_j} \mapsto 0$, $f_i \mapsto 0$ also (since it's a polynomial involving only $x_{i_1}, ..., x_{i_k}$, and with no constant term).

As a result, one has the following:

$$x_{i_k+1} = \sum_{i=1}^{n} g_i f_i \mapsto 0 \tag{2.2}$$

because each $f_i \mapsto 0$. However, this contradicts our assumption that $x_{i_k+1} \mapsto 1$. Hence, our assumption must be false, the ideal I cannot be a finitely generated R-module.

Hence, we found an exact sequence of R-modules $0 \to I \hookrightarrow R \twoheadrightarrow R/I \to 0$, such that the middle is finitely generated R-module, while the two sides are not guaranteed to be finitely generated (in particular, the left side is not guaranteed). This proves that the converse is in general false.

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Problem 3

Let M and N be flat R-modules, where R is a commutative ring. Prove or disprove that $M \otimes_R N$ is flat R-module.

Solution: We'll prove that $M \otimes_R N$ is also a flat R-module.

Since the tensor functor $(\_) \otimes_R (M \otimes_R N) : \text{R-Mod} \to \text{R-Mod}$ is a right exact functor, to show that $M \otimes_R N$ is flat (or the above tensor functor is exact), one only needs to prove it preserves injective R-linear maps.

Recall that for any R-module K, L, P, there is a unique R-linear isomorphism $\varphi_{KLP} : (K \otimes_R L) \otimes_R P \xrightarrow{\sim} K \otimes_R (L \otimes_R P)$ that satisfies $\varphi_{KLP}((k \otimes l) \otimes p) = k \otimes (l \otimes p)$ for all $k \in K$, $l \in L$, and $p \in P$.

Now, given an injective R-linear map $f : K \hookrightarrow K'$, since M is flat, then the exactness of the functor $(\_) \otimes_R M$ guarantees the map $f \otimes \mathrm{id}_M : K \otimes_R M \to K' \otimes_R M$ is injective (which the map is given by $(f \otimes \mathrm{id}_M)(k \otimes m) = f(k) \otimes m$ for all $k \in K$, $m \in M$).

Similarly, since N is flat, then the exactness of the functor $(\_) \otimes_R N$ guarantees the map $(f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N : (K \otimes_R M) \otimes_R N \to (K' \otimes_R M) \otimes_R N$ is injective (which the map is given by $((f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N)((k \otimes m) \otimes n) = (f(k) \otimes m) \otimes n$ for all $k \in K$, $m \in M$, and $n \in N$).

Finally, consider the following diagram:

$$\begin{tikzcd}
K \otimes_R (M \otimes_R N) \arrow[r, "f \otimes \mathrm{id}_{M \otimes_R N}"] \arrow[d, hook, two heads, "\varphi^{-1}_{KMN}"'] & K' \otimes_R (M \otimes_R N) \\
(K \otimes_R M) \otimes_R N \arrow[r, hook, "(f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N"'] & (K' \otimes_R M) \otimes_R N \arrow[u, hook, two heads, "\varphi_{K'MN}"']
\end{tikzcd}$$

Notice that the diagram commutes, because for each $k \in K$, $m \in M$, and $n \in N$, one has the following:

$$\left(f \otimes \mathrm{id}_{M \otimes_R N}\right)(k \otimes (m \otimes n)) = f(k) \otimes (m \otimes n) \tag{3.1}$$

$$\begin{aligned}
&\varphi_{K'MN} \circ ((f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N) \circ \varphi^{-1}_{KMN}(k \otimes (m \otimes n)) \\
&= \varphi_{K'MN} \circ ((f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N)((k \otimes m) \otimes n) \\
&= \varphi_{K'MN}((f(k) \otimes m) \otimes n) \\
&= f(k) \otimes (m \otimes n)
\end{aligned} \tag{3.2}$$

Then, since $K \otimes_R (M \otimes_R N)$ is generated by all elements of the form $k \otimes (m \otimes n)$, then the above two calculation shows that the two maps are the same, hence the diagram commutes.

As a result, since $\varphi_{K'MN}$, $(f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N$, and φ^{-1}_{KMN} are all injective, then their composition – which results in $f \otimes \mathrm{id}_{M \otimes_R N}$ – is injective. Therefore, the functor $(\_) \otimes_R (M \otimes_R N)$ preserves injective maps, hence is exact. This shows that $M \otimes_R N$ is a flat module.

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Problem 4

Let $\varphi : R \to S$ be a ring homomorphism and let M be a flat R-module. Prove or disprove that $M \otimes_R S$ is a flat S-module.

Solution: We'll prove that given M is a flat R-module, then $M \otimes_R S$ is a flat S-module.

Since $(M \otimes_R S) \otimes_S (\_) : \mathsf{S\text{-}Mod} \to \mathsf{S\text{-}Mod}$ is right exact, one just needs to prove it preserves injective maps.

For this, let's prove an exercise in Atiyah Macdonald Chapter 2:

Exercise: 2.15

Let A, B be commutative rings, let M be an A-module, P a B-module and N an (A, B)-bimodule (that is, N is simultaneously an A-module and B-module and the two structures are compatible in the sens that $a(xb) = (ax)b$ for all $a \in A$, $b \in B$, $x \in N$).

Then, $M \otimes_A N$ is naturally a B-module, $N \otimes_B P$ an A-module, and we have:

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P) \tag{4.1}$$

Proof: We'll prove the three claims in sequence:

I. B-Module Structure on $M \otimes_A N$:

For any $b \in B$, $m \in M$ and $n \in N$, detine the action $B \times (M \otimes_A N) \to M \otimes_A N$ as $b \cdot (m \otimes_A n) := m \otimes_A (nb)$ on each generator $m \otimes_A n$ (where nb is given by the B-module structure of N), we also define its action on any sum $\sum_{i=1}^{k} m_i \otimes_A n_i$ by sum of its action on individual $m_i \otimes_A n_i$. As a set map this is well-defined, let's check it does satisfy the B-module properties (which suffices to check for the generators of $M \otimes_A N$). Given any $b, b' \in B$, and any element $m \otimes_A n, m' \otimes_A n' \in M \otimes_A N$, we have the following:

1. $$\begin{aligned}(b + b') \cdot (m \otimes_A n) &= m \otimes_A (n(b + b')) = m \otimes_A (nb + nb') \\ &= m \otimes_A (nb) + m \otimes_A (nb') \\ &= b \cdot (m \otimes_A n) + b' \cdot (m \otimes_A n)\end{aligned} \tag{4.2}$$
2. $$b \cdot (m \otimes_A n + m' \otimes_A n') = b \cdot (m \otimes_A n) + b \cdot (m' \otimes_A n') \tag{4.3}$$

   (Note: This is by our definition, such that the action on the sum, is the sum of each individual generator after the action).
3. $$\begin{aligned}(bb') \cdot (m \otimes_A n) &= m \otimes_A (n(bb')) = m \otimes_A (n(b'b)) \\ &= m \otimes_A ((nb')b) = b \cdot (m \otimes_A (nb')) \\ &= b \cdot (b' \cdot (m \otimes_A n))\end{aligned} \tag{4.4}$$
4. $$1_B \cdot (m \otimes_A n) = m \otimes_A (n1_B) = m \otimes_A n \tag{4.5}$$

These are all the requirements for being a B-module, hence $M \otimes_A N$ has a B-module structure (by its action on N).

As a side note, this in fact makes $M \otimes_A N$ an (A, B)-bimodule structure, since for any $m \otimes_A n \in M \otimes_A N$, $a \in A$, and $b \in B$, we have the following:

$$\begin{aligned}a \cdot (b \cdot (m \otimes_A n)) &= a \cdot (m \otimes_A (nb)) = m \otimes_A (a(nb)) \\ &= m \otimes_A ((an)b) = b \cdot (m \otimes_A (an)) \\ &= b \cdot (a \cdot (m \otimes_A n))\end{aligned} \tag{4.6}$$

Which, the A-action and B-action commutes, showing the bimodule property.

II. A-Module Structure on $N \otimes_B P$:

Similarly, define $A \times (N \otimes_B P) \to N \otimes_B P$ by $a \cdot (n \otimes_B p) := (an) \otimes_B p$. Based on similar reasonings in **I**, it satisfies all the conditions for A-module, hence $N \otimes_B P$ now can be realized as A-module.

Again, similar to the proof in **I**, the computation shows that $N \otimes_B P$ is an (A, B)-bimodule.

III. Isomorphism:

We aim to show that $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are isomorphic as both A and B-module. For this, we'll construct a B-linear map $(M \otimes_A N) \otimes_B P \to M \otimes_A (N \otimes_B P)$, and another A-linear map $(M \otimes_A N) \otimes_B P \leftarrow M \otimes_A (N \otimes_B P)$, and prove that these two maps are isomorphism for abelian groups, while both maps are A-linear and B-linear, which will complete the isomorphism.

1. **B-linear map:**

   First, fix arbitrary $p \in P$, and define a map $F_p : M \times N \to M \otimes_A (N \otimes_B P)$ by $(m, n) \mapsto m \otimes_A (n \otimes_B p)$ for all $(m, n) \in M \times N$. Note that this is A-bilinear, since for all $a, a' \in A$, $m, m' \in M$, and $n, n' \in N$, we have the following:

$$\begin{aligned} F_p(am + a'm', n) &= (am + a'm') \otimes_A (n \otimes_B p) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m' \otimes_A (n \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m, n) \end{aligned} \tag{4.7}$$

$$\begin{aligned} F_p(m, an + a'n') &= m \otimes_A ((an + a'n') \otimes_B p) \\ &= m \otimes_A (a \cdot (n \otimes_B p) + a' \cdot (n' \otimes_B p)) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m \otimes_A (n \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m, n') \end{aligned} \tag{4.8}$$

   Hence, this uniquely factors through the tensor product, as $\overline{F}_p : M \otimes_A N \to M \otimes_A (N \otimes_B P)$ by $\overline{F}_p(m \otimes_A n) = m \otimes_A (n \otimes_B p)$.

   Now, let's define a map $G : (M \otimes_A N) \times P \to M \otimes_A (N \otimes_B P)$ as this:

$$G\left(\sum_{i=1}^{k} m_i \otimes_A n_i \quad , \quad p \right) := \overline{F}_p \left(\sum_{i=1}^{k} m_i \otimes_A n_i \right) \tag{4.9}$$

   We claim that G is B-bilinear (which suffices to check each individual tensors). Given any $b, b' \in B$, $(m \otimes_A n), (m' \otimes_A n') \in M \otimes_A N$, and $p, p' \in P$, we have the following:

$$\begin{aligned}
G(b\cdot(m\otimes_A n)+b'\cdot(m'\otimes_A n'),p) &= \overline{F}_p(m\otimes_A(nb)+m'\otimes_A(n'b'))\\
&= m\otimes_A((nb)\otimes_B p)+m'\otimes_A((n'b')\otimes_B p)\\
&= m\otimes_A((n\otimes_B p)b)+m'\otimes_A((n'\otimes_B p)b')\\
&= b\cdot(m\otimes_A(n\otimes_B p))+b'\cdot(m'\otimes_A(n'\otimes_B p))\\
&= b\cdot\overline{F}_p(m\otimes_A n)+b'\cdot\overline{F}_p(m'\otimes_A n')\\
&= b\cdot G(m\otimes_A n,p)+b'\cdot G(m'\otimes_A n',p)
\end{aligned}\tag{4.10}$$

$$\begin{aligned}
G(m\otimes_A n,bp+b'p') &= F_{bp+b'p'}(m\otimes_A n)\\
&= m\otimes_A(n\otimes_B(bp+b'p'))\\
&= m\otimes_A(b\cdot(n\otimes_B p)+b'\cdot(n\otimes_B p'))\\
&= b\cdot(m\otimes_A(n\otimes_B p))+b'\cdot(m\otimes_A(n\otimes_B p'))\\
&= b\cdot\overline{F}_p(m\otimes_A n)+b'\cdot\overline{F}_{p'}(m\otimes_A n)\\
&= b\cdot G(m\otimes_A n,p)+b'\cdot G(m\otimes_A n,p')
\end{aligned}\tag{4.11}$$

Since G is B-bilinear, there exists a unique B-linear map $\overline{G}:(M\otimes_A N)\otimes_B P\to M\otimes_A(N\otimes_B P)$ by $\overline{G}((m\otimes_A n)\otimes_B p)=m\otimes_A(n\otimes_B p)$.

2. **A-linear map:**

   Similarly, fix arbitrary $m\in M$, and define a map $H_m:N\times P\to(M\otimes_A N)\otimes_B P$ by $(n,p)\mapsto(m\otimes_A n)\otimes_B p$ for all $(n,p)\in N\times P$. Using the same logic in the previous part, one can prove that H_m is B-bilinear, hence inducing a unique B-linear map $\overline{H}_m: N\otimes_B P\to(M\otimes_A N)\otimes_B P$ by $\overline{H}_m(n\otimes_B p)=m\otimes_A(n\otimes_B p)$.

   Then, one defines another map $L:M\times(N\otimes_B P)\to(M\otimes_A N)\otimes_B P$ as follow:

$$L\left(m,\sum_{i=1}^k n_i\otimes_B p_i\right):=\overline{H}_m\left(\sum_{i=1}^k n_i\otimes_B p_i\right)\tag{4.12}$$

   Similar to the previous part, we in fact have L being A-bilinear after running through the computation. Hence, this induces a unique A-linear map $\overline{L}:M\otimes_A(N\otimes_B P)\to(M\otimes_A N)\otimes_B P$ by $\overline{L}(m\otimes_A(n\otimes_B p))=(m\otimes_A n)\otimes_B p$.

3. **Isomorphism:**

   Given the B-linear map $\overline{G}:(M\otimes_A N)\otimes_B P\to M\otimes_A(N\otimes_B P)$, and the A-linear map $\overline{L}:M\otimes_A(N\otimes_B P)\to(M\otimes_A N)\otimes_B P$, if we first look at the two modules' abelian group structure, one have this for all $m\in M,n\in N,p\in P$:

$$\begin{aligned}
\overline{L}\circ\overline{G}((m\otimes_A n)\otimes_B p)=\overline{L}(m\otimes_A(n\otimes_B p))=(m\otimes_A n)\otimes_B p\\
\overline{G}\circ\overline{L}(m\otimes_A(n\otimes_B p))=\overline{G}((m\otimes_A n)\otimes_B p)=m\otimes_A(n\otimes_B p)
\end{aligned}\tag{4.13}$$

   Hence, as homomorphism of abelian grouops, $\overline{G},\overline{L}$ are mutual inverses (since the compositions act as identity on all the generators of the two spaces, in the right composition order).

   Now, it suffices to check that $\overline{G},\overline{L}$ are in fact both A-linear and B-linear.

   - For $\overline{G}$, we've constructed it as a B-linear map, so it suffices to prove the A-linear structure (in particular, suffices to prove it preserves the A-action, since additivity is given by the fact that $\overline{G}$ is a homomorphism of abelian group). Given any $a\in A$, and $(m\otimes_A n)\otimes_B p\in(M\otimes_A N)\otimes_B P$, one has the following:

$$\begin{aligned}
\overline{G}(a\cdot((m\otimes_A n)\otimes_B p)) &= \overline{G}((a\cdot(m\otimes_A n))\otimes_B p) \\
&= \overline{G}(((am)\otimes_A n)\otimes_B p) \\
&= (am)\otimes_A (n\otimes_B p) \\
&= a\cdot(m\otimes_A (n\otimes_B p)) \\
&= a\cdot\overline{G}((m\otimes_A n)\otimes_B p)
\end{aligned} \tag{4.14}$$

(Note: Here we use the fact that $(M\otimes_A N)\otimes_B P$ has an A-module structure by letting A act on $M\otimes_A N$, which is proven in **I**, because M, N can be arbitrary).

Hence, $\overline{G}$ preserves A-action, which is A-linear.

- Similarly, for $\overline{L}$, it's constructed as an A-linear map, so it sufices to prove it preserves the B-action. Given any $b\in B$, and $m\otimes_A(n\otimes_B p)\in M\otimes_A(N\otimes_B P)$, one has the following:

$$\begin{aligned}
\overline{L}(b\cdot(m\otimes_A (n\otimes_B p))) &= \overline{L}(m\otimes_A (b\cdot(n\otimes_B p))) \\
&= \overline{L}(m\otimes_A (n\otimes_B (bp))) \\
&= (m\otimes_A n)\otimes_B (bp) \\
&= b\cdot((m\otimes_A n)\otimes_B p) \\
&= b\cdot\overline{L}(m\otimes_A (n\otimes_B p))
\end{aligned} \tag{4.15}$$

(Note: Again, we use the fact that $M\otimes_A(N\otimes_B P)$ has B-module structure by letting B act on $N\otimes_B P$, proven in **II**).

Hence, $\overline{L}$ preserves B-action, which is a B-linear map.

This realizes $\overline{G}, \overline{L}$ as mutual inverses as both A-linear and B-linear map, hence proving that $(M\otimes_A N)\otimes_B P\cong M\otimes_A(N\otimes_B P)$, as (A,B)-bimodule. In particular, an isomorphism (as both A and B-module) between them that's constructed above, has the following form:

$$\begin{aligned}
&\varphi_{MNP} : (M\otimes_A N)\otimes_B P \xrightarrow{\sim} M\otimes_A (N\otimes_B P) \\
&\varphi_{MNP}((m\otimes_A n)\otimes_B p) = m\otimes_A (n\otimes_B p), \quad \forall m\in M, n\in N, p\in P
\end{aligned} \tag{4.16}$$

□

Since S is an (R,S)-bimodule (where it's both an R-module and S-module, while any $r\in R$, $s,t\in S$ has $(r\cdot t)s = (\varphi(r)t)s = \varphi(r)(ts) = r\cdot(ts)$, where the dot $\cdot$ represents the R-action on S), then one has the following isomorphism, for any S-module P:

$$\varphi_{MSP} : (M\otimes_R S)\otimes_S P \xrightarrow{\sim} M\otimes_R (S\otimes_S P), \quad \varphi_{MSP}((m\otimes_R s)\otimes_S p) = m\otimes_R (s\otimes_S p) \tag{4.17}$$

Now, given any injective S-linear map $f: P\hookrightarrow P'$ between two S-modules, one simply has $(\mathrm{id}_S)\otimes_S f : S\otimes_B P\hookrightarrow S\otimes_B P'$ be injective (since tensor an S-module over S is isomorphic to the original module).

However, $(\mathrm{id}_S)\otimes_S f$ is also an injective R-linear map, since it's additive, and for all $r\in R$, $s\in S$, and $p\in P$, one has the following:

$$
\begin{aligned}
((\mathrm{id}_S)\otimes_S f)(r\cdot(s\otimes_B p)) &= ((\mathrm{id}_S)\otimes_S f)((r\cdot s)\otimes_B p)\\
&= (r\cdot s)\otimes_B f(p)\\
&= r\cdot(s\otimes_B f(p))\\
&= r\cdot((\mathrm{id}_S)\otimes_S f)(s\otimes_B p)
\end{aligned} \tag{4.18}
$$

Hence, with M being a flat R-module, the R-linear map $\mathrm{id}_M\otimes_R((\mathrm{id}_S)\otimes_S f): M\otimes_R(S\otimes_S P)\hookrightarrow M\otimes_R(S\otimes_S P')$ is also injective.

Similar to the previous one, we also have $\mathrm{id}_M\otimes_R((\mathrm{id}_S)\otimes_S f)$ being an injective S-linear map, since it's additive, for any $m\in M$, $s,t\in S$, and $p\in P$, one has the following:

$$
\begin{aligned}
(\mathrm{id}_M\otimes_R((\mathrm{id}_S)\otimes_S f))(s\cdot(m\otimes_R(t\otimes_S p))) &= (\mathrm{id}_M\otimes_R((\mathrm{id}_S)\otimes_S f))(m\otimes_R(s\cdot(t\otimes_S p)))\\
&= m\otimes_R(((\mathrm{id}_S)\otimes_S f)((st)\otimes_S p))\\
&= m\otimes_R((st)\otimes_S f(p))\\
&= m\otimes_R(s\cdot(t\otimes_S f(p)))\\
&= s\cdot(m\otimes_R(t\otimes_S f(p)))\\
&= s\cdot[(\mathrm{id}_M\otimes_R((\mathrm{id}_S)\otimes_S f))(m\otimes_R(t\otimes_S p))]
\end{aligned} \tag{4.19}
$$

For simplisity, let's define $\psi := \mathrm{id}_M\otimes_R((\mathrm{id}_S)\otimes_S f)$ as the injective S-linear map.

Finally, one has the following diagram:

$$
\begin{tikzcd}
(M\otimes_R S)\otimes_S P \arrow[r, "(\mathrm{id}_M\otimes_R S)\otimes_S f"] \arrow[d, hook, two heads, "\varphi_{MSP}"'] & (M\otimes_R S)\otimes_S P' \\
M\otimes_R(S\otimes_S P) \arrow[r, hook, "\psi"'] & M\otimes_R(S\otimes_S P') \arrow[u, hook, two heads, "\varphi^{-1}_{MSP'}"']
\end{tikzcd}
$$

Notice that this big diagram commutes, because for all $m\in M$, $s\in S$, and $p\in P$, one has the following:

$$
((\mathrm{id}_M\otimes_R S)\otimes_S f)((m\otimes_R s)\otimes_S p) = (m\otimes_R s)\otimes_S f(p) \tag{4.20}
$$

$$
\begin{aligned}
\varphi^{-1}_{MSP'}\circ\psi\circ\varphi_{MSP}((m\otimes_R s)\otimes_S p) &= \varphi^{-1}_{MSP'}\circ\psi(m\otimes_R(s\otimes_S p))\\
&= \varphi^{-1}_{MSP'}(m\otimes_R(s\otimes_S f(p)))\\
&= (m\otimes_R s)\otimes_S f(p)
\end{aligned} \tag{4.21}
$$

This shows that $\varphi^{-1}_{MSP'}\circ\psi\circ\varphi_{MSP} = (\mathrm{id}_M\otimes_R S)\otimes_S f$ as S-linear maps. Hence, with each map on the LRS being injective, the RHS is also injective.

This shows that $(M\otimes_R S)\otimes_S (\_)$ is an exact functor (as it preserves injectivity), hence $M\otimes_R S$ can be realized as a flat S-module.

5 D

Problem 5

Let $\varphi : R \to S$ be a ring homomorphism such that M is projective as an R-module. Prove or disprove that $M \otimes_R S$ is projective as S-module.

Solution: We'll prove that if M is projective as R-module, then $M \otimes_R S$ is projective as S-module.

Since M is projective as an R-module, there exists another R-module N, such that $M \oplus N \cong \bigoplus_{i\in I} Re_i$ a free R-module (where each $Re_i \cong R$ as R-module). Then, since tensor product „commutes“ with direct sum (up to isomorphism), hence one has the following isomorphism as R-modules:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong (M \oplus N) \otimes_R S \cong \left(\bigoplus_{i\in I} Re_i\right) \otimes_R S \cong \bigoplus_{i\in I}(Re_i \otimes_R S) \tag{5.1}$$

As a remark, all these isomorphism respects the S-module structure: Given any family of R-modules $\{M_i\}_{i\in\Lambda}$, the R-linear isomorphism $\varphi : \left(\bigoplus_{i\in\Lambda} M_i\right) \otimes_R S \xrightarrow{\sim} \bigoplus_{i\in I}(M_i \otimes_R S)$ is given as:

$$\varphi\Big(\big((m_i)_{i\in\Lambda}\big) \otimes s\Big) = (m_i \otimes s)_{i\in\Lambda} \tag{5.2}$$

Which, this map is also S-linear, as any $(m_i)_{i\in\Lambda} \in \bigoplus_{i\in\Lambda} M_i$, and $s, t \in S$, satisfy the following:

$$\begin{aligned}\varphi\Big(s \cdot \big((m_i)_{i\in\Lambda} \otimes t\big)\Big) &= \varphi\Big((m_i)_{i\in\Lambda} \otimes (ts)\Big) = (m_i \otimes (ts))_{i\in\Lambda} \\ &= (s \cdot (m_i \otimes t))_{i\in\Lambda} = s \cdot (m_i \otimes t)_{i\in\Lambda}\end{aligned} \tag{5.3}$$

Hence, φ is also an S-linear isomorphism. Apply this logic to the 1st and 3rd isomorphism relation, they're isomorphic as S-module (the middle part is true because S is preserved).

Now, look at each $Re_i \cong R$, one has each $Re_i \otimes_R S \cong S$ as R-module (and a specific R-linear isomorphism is given by $\varphi : R \otimes_R S \xrightarrow{\sim} S$, $\varphi(r \otimes s) = r \cdot s = \varphi(r)s$).

However, notice that this φ is also an S-linear map, because it's additive, and also every $r \in R$ and $s, t \in S$ satisfy the following:

$$\varphi(s \cdot (r \otimes t)) = \varphi(r \otimes (ts)) = r \cdot (ts) = \varphi(r)(ts) = (\varphi(r)t)s = s \cdot (r \cdot t) = s \cdot \varphi(r \otimes t) \tag{5.4}$$

(Note: Here the S-module structure on $R \otimes_R S$ is by letting S act on S).

Hence, it's not only an R-linear isomorphism, but in fact an S-linear isomorphism. As a consequence, with the S-module structure on $R \otimes_R S$ being isomorphic to S, one has:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong \bigoplus_{i\in I}(Re_i \otimes_R S) \cong \bigoplus_{i\in I} S \tag{5.5}$$

Where the final part is a free S-module. This realizes $M \otimes_R S$ as a direct summand of a free S-module, hence it's projective as S-module.

6 ND

Problem 6

Let $\varphi : R \to S$ be a ring homomorphism and let M be an injective R-module such that $M \otimes_R S \neq 0$. Prove or disprove that $M \otimes_R S$ is injective S-module.

Solution: Guess: $R = \mathbb{Z}$, $S = \mathbb{Z}_{(p)}$ for some prime p, choose $M = \mathbb{Q}$, then $M \otimes_R S \neq 0$ I believe (even though it's fxxking hard to check). I think $M \otimes_R S \cong M$ as S-module in this case (bc the description of $\mathbb{Z}_{(p)}$ can be realized as a subring of $\mathbb{Q}$, and one can simply move all coefficients into $\mathbb{Q}$ by divisibility of $\mathbb{Q}$).

Is $\mathbb{Q}$ an injective S-mod? Yes, bc the fraction field of S is $\mathbb{Q}$, and S is a PID.

Maybe it's true then...

7 HD (Type it up)

Problem 7

Let $\varphi : R \to S$ be a ring homomorphism such that S is flat R-algebra. Prove or disprove that φ is injective.

Solution: We'll disprove it by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, recall that it has a projection $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ by $\varphi(\overline{1}_6) = \overline{1}_3$, this realizes $\mathbb{Z}/3\mathbb{Z}$ as an R-algebra, while the map φ is not injective (since $\varphi(\overline{3}_6) = \overline{3}_3 = 0$).

(Basically, the idea is that as R-modules, $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ because the section $\mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ by $\overline{1}_3 \mapsto \overline{2}_6$ splits, as a result $S = \mathbb{Z}/3\mathbb{Z}$ is a projective R-mod, hence flat)..

8 ND

Problem 8

Let $\varphi : R \to S$ be an injective ring homomorphism such that S is a field. Prove or disprove that S is a flat R-algebra.

Solution: Notice that R is a subring of a field, hence an integral domain. Given K as fraction field of R, one realizes S as a field extension of K, hence $S \cong \bigoplus_{i \in I} K$ as K-vector space. Notice that they're also isomorphic as R-modules (simply because a K-linear map is also an R-linear map, by restricting the scalars). So, it suffices to argue that K is flat or not (since direct sum commutes with tensor).

If R is a field then we're done (everything is flat).

If R is not a field, how to characterize flatness? We know K definitely can't be free, and there are cases where it's not projective (EX: $\mathbb{Q}$ over $\mathbb{Z}_{(p)}$ covered in class).

9 ND

Problem 9

> Let R be a dedekind domain and let $I \subset R$ be an ideal. Prove or disprove that I is a flat R-module.

Solution:

10 ND

Problem 10

> Let R be a PID and let M be a finitely generated flat R-module. Prove or disprove that M is a free R-module.

Solution: Q: Over a PID, is there any flat module that's not projective?

Guess: $k[x]$ as a $k[x^2]$-module (which $k[x^2]$ is a PID, since it's iso to $k[y]$). Then, as a module it's generated by 1 and x (since separate out the odd degree terms and even degree terms we're good). Oh shit but it's then free.

What happens if we consider minimal generating set? Say $x_1, ..., x_n$ such that it's minimal, we find a surjection $F_R(x_1, ..., x_n) \twoheadrightarrow M$.

11 HD (Type it up)

Problem 11

> Let R be a PID and let M be an R-module. Prove that M is a submodule of an injective R-module.

Solution: Consider K as the fraction field of R, then any $\bigoplus_{i\in I} K$ is injective (because it's divisible coordinate wise!)

For any M an R-module, there exists free R-module $\bigoplus_{i\in J} R$ that surjects onto M, hence $M \cong \left(\bigoplus_{i\in J} R\right)/L$ for some submodule L of the free module.

Now, consider the inclusion of R-modules $L \hookrightarrow \bigoplus_{i\in J} R \hookrightarrow \bigoplus_{i\in J} K$, and the quotient $\bigoplus_{i\in J} K \twoheadrightarrow \left(\bigoplus_{i\in J} K\right)/L$. Then, then map $\bigoplus_{i\in J} R \to \left(\bigoplus_{i\in J} K\right)/L$ has kernel precisely L, showing $\left(\bigoplus_{i\in J} R\right)/L \hookrightarrow \left(\bigoplus_{i\in J} K\right)/L$ by its natural identification.

So, M injects into $\left(\bigoplus_{i\in J} K\right)/L$, which is divisible (since it's a module theoretic quotient of $\bigoplus_{i\in J} K$ which is divisible).

12 HD (type it up)

Problem 12

> Let R be a commutative ring and let m be a given maximal ideal of R. Suppose m is an R-module such that $M_m = 0$. Prove or disprove that $M = 0$.

Solution: I guess we can take $R = \mathbb{Z}$, $m = 2\mathbb{Z}$, and $M = \mathbb{Z}/3\mathbb{Z}$. Then, M_m denotes all the elements of the form $\frac{a_3}{d}$, where $d \notin 2\mathbb{Z}$ (i.e. we can divide by 3!) So in particular, one has the following:

$$3 \cdot \left(1 \cdot \overline{a}_3 - 1 \cdot \overline{0}_3\right) = 3 \cdot \overline{a}_3 = \overline{0}_3 \in M \tag{12.1}$$

This shows that $\frac{\overline{a}_3}{1} = \frac{0}{1}$ in M_m, showing $M_m = 0$, while $M \neq 0$.
(But if swapping to all maximal ideal m I think it's true).

13 ND

Problem 13

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be two ring homomorphisms such that S is R-flat and T is S-flat. Prove or disprove that T is R-flat.

Solution:

14 ND

Problem 14

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R-flat. Prove or disprove that φ can not be surjective.

Solution:

15 ND

Problem 15

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R-flat. Prove or disprove that ψ can not be surjective.

Solution:

16 ND

Problem 16

Let $R \xrightarrow{\varphi} S$ be ring homomorphism which is flat. Let M be an R-module such that $M \otimes_R S$ is torsion-free S-module. Is M torsion-free R-module?

Solution: The counterexample I can thought of are all not flat. I want to say it's true.